Question	Answers	Extra information	Mark	AO / Spec Ref.
03.4	any **two** from: • fewer lynx (to eat them) • more food / grass (available) • hares reproduce / breed / multiply	allow not many predators / lynx do **not** accept no lynx / predators allow a lot of food / grass	2	AO2 4.7.1.3 4.7.2.1

Question	Answers	Extra information	Mark	AO / Spec Ref.
03.5	(number of lynx) increases		1	AO3 4.7.1.3 4.7.2.1

Question	Answers	Extra information	Mark	AO / Spec Ref.
03.6	less food **or** fewer (snowshoe) hares (to eat) **or** fewer prey	ignore the lynx were hunted do **not** accept no food	1	AO2 4.7.1.3 4.7.2.1

Question	Answers	Extra information	Mark	AO / Spec Ref.
03.7	any **two** from: • (lost in) respiration (of snowshoe hare) • egestion / faeces • excretion / urea / urine • not all eaten	allow as carbon dioxide allow not all digested allow not all (named) parts eaten if no other mark awarded, allow **1** mark for waste	2	AO1 4.7.2.1 4.7.4.3

Total Question 3	**13**

Question 4

Question	Answers	Extra information	Mark	AO / Spec Ref.
04.1	6000 (kJ/m^2/year)		1	AO2 4.7.5.2

Question	Answers	Extra information	Mark	AO / Spec Ref.
04.2	$\frac{40}{16\,000} \times 100$ 0.25 (%)		1 1	AO2 4.7.5.2

Question	Answers	Extra information	Mark	AO / Spec Ref.
04.3	cows move less indoors it is warmer indoors		1 1	AO1 4.7.5.2

Question	Answers	Extra information	Mark	AO / Spec Ref.
04.4	cows are close(r) together	allow cows are in an enclosed space	1	AO2 4.7.5.2 4.3.1.1

Question	Answers	Extra information	Mark	AO / Spec Ref.
04.5	any **one** from: • D • G • H		1	AO3 4.6.3.7 4.1.1.6 4.3.1.8

Question	Answers	Extra information	Mark	AO / Spec Ref.
04.6	DNA		1	AO1 4.6.3.7

Question	Answers	Extra information	Mark	AO / Spec Ref.
04.7	reproduction		1	AO2 4.6.3.7 4.3.1.1

Question	Answers	Extra information	Mark	AO / Spec Ref.
04.8	any **two** from: • many animals will become ill • resistant bacteria passed from animal to animal • animals cannot be cured (by the antibiotic) • animals may be less productive / efficient • farmer may lose profit / money	ignore reference to crops allow many animals may die allow disease passed from animal to animal **or** other animals infected allow antibiotic will not work allow less meat / milk produced allow less growth allow cannot sell infected produce / animals	2	AO2 4.3.1.1 4.6.3.7 4.7.5.2

Total Question 4	**11**

Question 5

Question	Answers	Extra information	Mark	AO / Spec Ref.
05.1	fungi		1	AO1 4.7.2.3

Question	Answers	Extra information	Mark	AO / Spec Ref.
05.2	the pH of the milk		1	AO2 4.7.2.3 RPA10

Question	Answers	Extra information	Mark	AO / Spec Ref.
05.3	put the beaker in a water bath		1	AO2 4.7.2.3 RPA10

Question	Answers	Extra information	Mark	AO / Spec Ref.
05.4	fatty acid		1	AO1 4.2.2.1 4.7.2.3 RPA10

Question	Answers	Extra information	Mark	AO / Spec Ref.
05.5	all the fat had been digested		1	AO2 4.7.2.3 4.2.2.1 RPA10

Question	Answers	Extra information	Mark	AO / Spec Ref.
05.6	any **one** from: • (there is) more (kinetic) energy • enzyme activity is higher • bacteria / microorganisms are dividing / reproducing faster	allow converse if clearly describing 5 °C allow particles move faster allow more collisions between particles allow enzymes work faster ignore enzymes work better allow number of bacteria / microorganisms increasing faster allow more bacteria / microorganisms ignore bacteria / microorganisms grow faster ignore it is warmer	1	AO2 4.7.2.3 4.1.1.6 4.2.2.1 RPA10

Question	Answers	Extra information	Mark	AO / Spec Ref.
05.7	(30 °C, 2 days) 6(.0) **and** (30 °C, 3 days) 4.7 (fall) 1.3 (pH units/day)	allow a tolerance of ±½ small square allow −1.3 (pH units/day) allow correct answer using student's incorrect readings in the range of 5.95 to 6.05 and / or 4.65 to 4.75	1 1	AO2 4.7.2.3 4.2.2.1 RPA10

Question	Answers	Extra information	Mark	AO / Spec Ref.
05.8	$\frac{1.3}{0.1}$ 13	allow ecf from answer in question **05.7** do **not** accept if a unit is given	1 1	AO2 4.7.2.3 4.2.2.1 RPA10

Total Question 5	**10**

Question 6

Question	Answers	Mark	AO / Spec Ref.
06.1	**Substance** / **Reason** Glucose → The molecules are small enough to pass through the filter Protein → The molecules are too large to pass through the filter The molecules are too small to pass through the filter do **not** accept more than one line from a box on the left	1 1	AO3 4.5.3.3

Question	Answers	Extra information	Mark	AO / Spec Ref.
06.2	158.1 (dm^3)		1	AO2 4.5.3.3